

Schrödinger's Equation: A Wave Equation or Heat Equation?

1 Introduction

Schrödinger's equation is central to quantum mechanics and is often introduced as a wave equation, describing particles in the same way classical wave equations describe vibrating strings or light. This idea stems from de Broglie's hypothesis, which proposed that particles exhibit wave-like properties, with wavelengths inversely proportional to their momentum.

Building on this, Erwin Schrödinger formulated a differential equation to describe how quantum systems evolve over time, successfully capturing interference, tunneling, and quantized energy levels.

But is Schrödinger's equation really a wave equation? Or could it be more like something else, such as the diffusion equation that governs the flow of heat?

A TALE OF TWO EQUATIONS

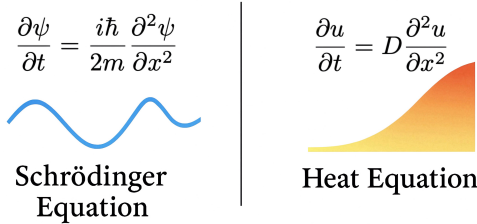


Figure 1: A comparison between the time-dependent Schrödinger equation for a free particle (left) and the classical heat equation with real diffusion constant D (right).

2 Relation to Lecture Course

In Lecture 2, Schrödinger's equation was introduced as a natural consequence of combining de Broglie's relations with a dynamical equation for wavefunctions. The time-dependent Schrödinger equation for a particle of mass m in one dimension, subject to a potential $V(x)$, is:

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x, t) \quad (1)$$

This equation determines how a quantum state evolves over time and is the foundation for much of the quantum theory studied throughout the course.

3 The Diffusion Analogy

The heat (or diffusion) equation, which describes how temperature evolves in one dimension over time, is:

$$\frac{\partial u(x, t)}{\partial t} = D \frac{\partial^2 u(x, t)}{\partial x^2} \quad (2)$$

where D is a real, positive constant known as the diffusion coefficient.

If, in equation (1), we set the potential $V(x) = 0$, the equation simplifies exactly to the form of the heat equation (2), with one crucial difference,

$$D = \frac{i\hbar}{2m} \quad (3)$$

In this version, the diffusion constant becomes imaginary rather than real. This surprising resemblance between the two equations suggests that Schrödinger's equation can be interpreted as a heat equation with an imaginary diffusion constant.

To understand what this really means, we can examine the form of the solutions. In the classical heat equation. The time-dependent part of the solution to the heat equation is typically:

$$u(t) \propto e^{-Dk^2t} \quad (4)$$

which describes exponential decay. This makes sense for temperature, it evens out over time as thermal energy diffuses.

But if we substitute the imaginary diffusion constant from equation (3), we obtain:

$$\psi(t) \propto e^{-i\omega t} \quad (5)$$

where $\omega = \frac{\hbar k^2}{2m}$. This function oscillates instead of decaying. It represents a rotation in the complex plane, as described by Euler's identity:

$$e^{-i\omega t} = \cos(\omega t) - i \sin(\omega t) \quad (6)$$

Instead of flattening out, the wavefunction exhibits sustained oscillations in phase.

This analogy provides an intuitive way to visualise quantum evolution by imagining wavefunctions diffusing over imaginary time. More importantly, it highlights why complex numbers are essential in quantum mechanics, they allow wavefunctions to evolve through pure phase rotations (6) rather than exponential decay. This oscillatory behavior preserves the wavefunction's phase information, enabling interference and superposition, key features absent from real-valued equations like the heat equation. Thus, the subtle introduction of an imaginary diffusion constant transforms a classical diffusion process into the coherent, non-classical evolution characteristic of quantum systems.

References

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